

ROW_NUMBER

RANK

DENSE_RANK

lead

lag

drop table employees;

```
CREATE TABLE employees (  
  employee_id INT PRIMARY KEY,  
  employee_name VARCHAR(50),  
  department VARCHAR(50),  
  salary DECIMAL(10, 2),  
  hire_date DATE  
);
```

```
INSERT INTO employees (employee_id, employee_name, department, salary, hire_date)  
VALUES
```

```
(1, 'Amit', 'HR', 50000, '2022-01-15'),  
(2, 'Neha', 'HR', 55000, '2023-03-10'),  
(3, 'Suresh', 'HR', 48000, '2021-11-20'),  
(4, 'Rohit', 'HR', 52000, '2022-09-05'),  
(5, 'Raj', 'Finance', 60000, '2021-07-23'),  
(6, 'Ravi', 'Finance', 62000, '2022-09-01'),  
(7, 'Kiran', 'Finance', 58000, '2021-02-14'),  
(8, 'Sunita', 'Finance', 61000, '2023-01-11'),  
(9, 'Priya', 'IT', 70000, '2020-12-02'),  
(10, 'Anjali', 'IT', 67000, '2021-11-19'),  
(11, 'Vikas', 'IT', 69000, '2022-05-20'),  
(12, 'Sanjay', 'IT', 72000, '2023-04-30'),  
(13, 'Meena', 'IT', 68000, '2021-03-15');
```

```
select employee_id,  
employee_name,  
department,  
salary,  
hire_date,  
row_number() over (partition by department order by salary desc) as row_num  
from employees;
```

but what if we have a requirement to find total salary paid for each department

ideally how you approach this based on whatever you have learnt till now?

dept_agg

HR , 241000

Finanace, 205000

IT, 250000

employees

```
(1, 'Amit', 'HR', 50000, '2022-01-15'),
(2, 'Neha', 'HR', 55000, '2023-03-10'),
(3, 'Suresh', 'HR', 48000, '2021-11-20'),
(4, 'Rohit', 'HR', 52000, '2022-09-05'),
(5, 'Raj', 'Finance', 60000, '2021-07-23'),
(6, 'Ravi', 'Finance', 62000, '2022-09-01'),
(7, 'Kiran', 'Finance', 58000, '2021-02-14'),
(8, 'Sunita', 'Finance', 61000, '2023-01-11'),
(9, 'Priya', 'IT', 70000, '2020-12-02'),
(10, 'Anjali', 'IT', 67000, '2021-11-19'),
(11, 'Vikas', 'IT', 69000, '2022-05-20'),
(12, 'Sanjay', 'IT', 72000, '2023-04-30'),
(13, 'Meena', 'IT', 68000, '2021-03-15');
```

```
select e.*,d.total_sal
from employees e
join
(
select department, sum(salary) as total_sal
from employees
group by department
) d
on e.department = d.department;
```

can we achieve the above without using joins?

row_number
rank
dense_rank
lead
lag

we can do it using window functions

we can use aggregations as part of window functions

```
select *,
sum(salary) over (partition by department) as total_sal
from employees
```

```
-- row_number() over (partition by department order by salary desc)
-- sum(salary) over (partition by department)
```

here partition by is optional
now order by is also optional

```
CREATE TABLE employees (  
employee_id INT PRIMARY KEY,  
employee_name VARCHAR(50),  
department VARCHAR(50),  
city VARCHAR(50),  
salary DECIMAL(10, 2),  
hire_date DATE  
);
```

```
INSERT INTO employees (employee_id, employee_name, department, city, salary, hire_date)  
VALUES  
(1, 'Amit', 'HR', 'Mumbai', 50000, '2022-01-15'),  
(2, 'Neha', 'HR', 'Mumbai', 55000, '2023-03-10'),  
(3, 'Suresh', 'HR', 'Delhi', 48000, '2021-11-20'),  
(4, 'Rohit', 'HR', 'Delhi', 52000, '2022-09-05'),  
(5, 'Raj', 'Finance', 'Mumbai', 60000, '2021-07-23'),  
(6, 'Ravi', 'Finance', 'Delhi', 62000, '2022-09-01'),  
(7, 'Kiran', 'Finance', 'Mumbai', 58000, '2021-02-14'),  
(8, 'Sunita', 'Finance', 'Delhi', 61000, '2023-01-11'),  
(9, 'Priya', 'IT', 'Mumbai', 70000, '2020-12-02'),  
(10, 'Anjali', 'IT', 'Delhi', 67000, '2021-11-19'),  
(11, 'Vikas', 'IT', 'Mumbai', 69000, '2022-05-20'),  
(12, 'Sanjay', 'IT', 'Delhi', 72000, '2023-04-30'),  
(13, 'Meena', 'IT', 'Delhi', 68000, '2021-03-15');
```

-- total salary for each city

```
select *,  
sum(salary) over (partition by city) as total_sal  
from employees  
select city, sum(salary)  
from employees  
group by city
```

-- total salary for each department in each city

```
select *,  
sum(salary) over (partition by city, department) as total_sal  
from employees
```

```
select *,
sum(salary) over (partition by department) as total_sal,
min(salary) over (partition by department) as min_sal,
max(salary) over (partition by department) as max_sal,
avg(salary) over (partition by department) as avg_sal,
count(salary) over (partition by department) as department_count
from employees
```

```
select *,
sum(salary) over () as total_sal,
min(salary) over () as min_sal,
max(salary) over () as max_sal,
avg(salary) over () as avg_sal,
count(salary) over () as department_count
from employees
```